ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 6 (PAPER 2) · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics practice exam 6 (paper 2).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Breadboard layouts may look different and still be correct: check that each connection in the circuit is made. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

Placement: 1 mark for each component placed on the breadboard ✓✓✓✓✓✓ (battery, LDR, VR1, R1, R2, Q1, LED) = 7. Polarity: battery ✓, LED (anode towards R2) ✓, transistor pins (collector, base, emitter) correct ✓ = 3. Connections, 1 mark each ✓✓✓✓✓✓✓✓✓: battery + and − to the supply rails; one end of VR1 to +9 V; VR1 wiper / other end joined to the LDR (the divider point); other end of the LDR to 0 V; divider point to R1; R1 to the base; R2 from +9 V to the LED anode; LED cathode to the collector; emitter to 0 V = 9. Name / examination number and printout ✓ = 1. (20)

[20]

QUESTION 2

Placement: IC across the centre gap of the breadboard ✓; S1 ✓; S2 ✓; R1 ✓; R2 ✓; R3 ✓; LED ✓; 5 V supply ✓ = 8. Polarity: IC the right way round (notch / pin 1 marked) ✓; LED ✓; supply ✓ = 3. Connections, 1 mark each ✓✓✓✓✓✓✓✓✓: pin 14 to +5 V; pin 7 to 0 V; S1 between +5 V and pin 1; S2 between +5 V and pin 2; R1 from pin 1 to 0 V; R2 from pin 2 to 0 V; pin 3 to R3; R3 to the LED anode; LED cathode to 0 V = 9. (20)

[20]

QUESTION 3

Correct circuit: probe 1 to +9 V; probe 2 through the 10 kΩ resistor to the base; emitter to 0 V; buzzer (+) to +9 V and buzzer (−) to the collector; LED anode through the 470 Ω resistor to +9 V and LED cathode to the collector (buzzer and LED branch in parallel).

Marks for each part, 2 marks each: 9 V battery with correct polarity ✓✓; two probes, one to +9 V ✓✓; BC547 with pins correct ✓✓; 10 kΩ resistor from probe to base ✓✓; buzzer in the collector circuit with correct polarity ✓✓; red LED with correct polarity ✓✓; 470 Ω resistor in series with the LED ✓✓; buzzer and LED branch in parallel, emitter to 0 V ✓✓; breadboard and jumpers used neatly ✓✓; printout with name / examination number ✓✓ (20)

[20]

QUESTION 4

4.1 12 V supply with polarity ✓; potentiometer across the supply with its wiper through the 1 kΩ resistor to the TIP120 base ✓; motor between +12 V and the collector, emitter to 0 V ✓; diode across the motor with its cathode to +12 V (reverse biased) ✓; LED with 1 kΩ resistor across the supply; correct symbols ✓ (5)

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4.2 Placement ✓✓✓✓✓✓✓ (supply, motor, TIP120, potentiometer, 1 kΩ base resistor, diode, LED, LED resistor) = 8. Polarity ✓✓✓: diode stripe towards +12 V; LED; TIP120 pins (base, collector, emitter) = 3. Connections ✓✓✓✓: potentiometer ends to +12 V and 0 V; wiper to the base resistor and on to the base; motor between +12 V and the collector; emitter to 0 V = 4. (15)

[20]

TOTAL: 80

END OF MARKING GUIDELINE